

Experiment 13: Host Discovery

Aim

To discover live hosts on a network using Nmap host discovery techniques.

Procedure

1. Ping Scan (discover live hosts without port scan):
`nmap -sn 192.168.1.0/24`
2. ARP Ping Scan (very reliable on local network):
`nmap -PR 192.168.1.0/24`
3. ICMP Echo Request discovery:
`nmap -PE 192.168.1.0/24`
4. Discover hosts and show only live ones:
`nmap -sn -T4 192.168.1.1-50`

Sample Output / Results

```
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.94 ...
Nmap scan report for 192.168.1.1
Host is up (0.0023s latency).
Nmap scan report for 192.168.1.10
Host is up (0.0041s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.85 seconds
```



kali@kali: ~

Session Actions Edit View Help

```
zsh: corrupt history file /home/kali/.zsh_history state UNKNOWN group default glen l
(kali@kali)-[~]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 12:30 -0400
Nmap scan report for 192.168.1.0
Host is up (0.047s latency).
Nmap scan report for 192.168.1.1
Host is up (0.027s latency).
Nmap scan report for 192.168.1.2
Host is up (0.10s latency).
Nmap scan report for 192.168.1.3
Host is up (0.080s latency).
Nmap scan report for 192.168.1.4
Host is up (0.024s latency).
Nmap scan report for 192.168.1.5
Host is up (0.058s latency).
Nmap scan report for 192.168.1.6
Host is up (0.058s latency).
Nmap scan report for 192.168.1.7
Host is up (0.058s latency).
Nmap scan report for 192.168.1.8
Host is up (0.10s latency).
Nmap scan report for 192.168.1.9
Host is up (0.10s latency).
Nmap scan report for 192.168.1.10
Host is up (0.11s latency).
Nmap scan report for 192.168.1.11
Host is up (0.057s latency).
Nmap scan report for 192.168.1.12
Host is up (0.031s latency).
Nmap scan report for 192.168.1.13
Host is up (0.055s latency).
Nmap scan report for 192.168.1.14
Host is up (0.055s latency).
Nmap scan report for 192.168.1.15
Host is up (0.055s latency).
Nmap scan report for 192.168.1.16
Host is up (0.058s latency).
Nmap scan report for 192.168.1.17
Host is up (0.10s latency).
Nmap scan report for 192.168.1.18
Host is up (0.10s latency).
Nmap scan report for 192.168.1.19
Host is up (0.10s latency).
Nmap scan report for 192.168.1.20
Host is up (0.089s latency).
Nmap scan report for 192.168.1.21
Host is up (0.091s latency).
Nmap scan report for 192.168.1.22
Host is up (0.098s latency).
Nmap scan report for 192.168.1.23
```



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Session Actions Edit View Help

```
Nmap scan report for 192.168.1.26: 45576:rdisc:nodev:state:UNKNOWN:3
Host is up (0.046s latency).
Nmap scan report for 192.168.1.27:
Host is up (0.028s latency).
Nmap scan report for 192.168.1.28:
Host is up (0.066s latency).
Nmap scan report for 192.168.1.29:
Host is up (0.064s latency).
Nmap scan report for 192.168.1.30:
Host is up (0.11s latency).
Nmap scan report for 192.168.1.31:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.32:
Host is up (0.068s latency).
Nmap scan report for 192.168.1.33:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.34:
Host is up (0.066s latency).
Nmap scan report for 192.168.1.35:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.36:
Host is up (0.068s latency).
Nmap scan report for 192.168.1.37:
Host is up (0.068s latency).
Nmap scan report for 192.168.1.38:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.39:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.40:
Host is up (0.070s latency).
Nmap scan report for 192.168.1.41:
Host is up (0.070s latency).
Nmap scan report for 192.168.1.42:
Host is up (0.073s latency).
Nmap scan report for 192.168.1.43:
Host is up (0.047s latency).
Nmap scan report for 192.168.1.44:
Host is up (0.028s latency).
Nmap scan report for 192.168.1.45:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.46:
Host is up (0.11s latency).
Nmap scan report for 192.168.1.47:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.48:
Host is up (0.070s latency).
Nmap scan report for 192.168.1.49:
Host is up (0.070s latency).
Nmap scan report for 192.168.1.50:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.51:
```



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Session Actions Edit View Help

```
Nmap scan report for 192.168.1.57: 65535 open ports, no queue state UNKNOW
Host is up (0.070s latency).
Nmap scan report for 192.168.1.58:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.59:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.60:
Host is up (0.093s latency).
Nmap scan report for 192.168.1.61:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.62:
Host is up (0.078s latency).
Nmap scan report for 192.168.1.63:
Host is up (0.100s latency).
Nmap scan report for 192.168.1.64:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.65:
Host is up (0.099s latency).
Nmap scan report for 192.168.1.66:
Host is up (0.020s latency).
Nmap scan report for 192.168.1.67:
Host is up (0.084s latency).
Nmap scan report for 192.168.1.68:
Host is up (0.084s latency).
Nmap scan report for 192.168.1.69:
Host is up (0.067s latency).
Nmap scan report for 192.168.1.70:
Host is up (0.047s latency).
Nmap scan report for 192.168.1.71:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.72:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.73:
Host is up (0.080s latency).
Nmap scan report for 192.168.1.74:
Host is up (0.064s latency).
Nmap scan report for 192.168.1.75:
Host is up (0.031s latency).
Nmap scan report for 192.168.1.76:
Host is up (0.10s latency).
Nmap scan report for 192.168.1.77:
Host is up (0.031s latency).
Nmap scan report for 192.168.1.78:
Host is up (0.031s latency).
Nmap scan report for 192.168.1.79:
Host is up (0.089s latency).
Nmap scan report for 192.168.1.80:
Host is up (0.081s latency).
Nmap scan report for 192.168.1.81:
Host is up (0.032s latency).
Nmap scan report for 192.168.1.82:
```




kali@kali: ~

Session Actions Edit View Help

```
Host is up (0.047s latency).
Nmap scan report for 192.168.1.237
Host is up (0.048s latency).
Nmap scan report for 192.168.1.238
Host is up (0.036s latency).
Nmap scan report for 192.168.1.239
Host is up (0.048s latency).
Nmap scan report for 192.168.1.240
Host is up (0.038s latency).
Nmap scan report for 192.168.1.241
Host is up (0.077s latency).
Nmap scan report for 192.168.1.242
Host is up (0.077s latency).
Nmap scan report for 192.168.1.243
Host is up (0.11s latency).
Nmap scan report for 192.168.1.244
Host is up (0.077s latency).
Nmap scan report for 192.168.1.245
Host is up (0.077s latency).
Nmap scan report for 192.168.1.246
Host is up (0.11s latency).
Nmap scan report for 192.168.1.247
Host is up (0.051s latency).
Nmap scan report for 192.168.1.248
Host is up (0.038s latency).
Nmap scan report for 192.168.1.249
Host is up (0.11s latency).
Nmap scan report for 192.168.1.250
Host is up (0.083s latency).
Nmap scan report for 192.168.1.251
Host is up (0.038s latency).
Nmap scan report for 192.168.1.252
Host is up (0.051s latency).
Nmap scan report for 192.168.1.253
Host is up (0.030s latency).
Nmap scan report for 192.168.1.254
Host is up (0.038s latency).
Nmap scan report for 192.168.1.255
Host is up (0.038s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 63.04 seconds
```



(kali@kali)-[~]



\$ ping -c 4 192.168.1.10

PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.

— 192.168.1.10 ping statistics —

4 packets transmitted, 0 received, 100% packet loss, time 3080ms